Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA:

Resource Allocation, User Capacity and Mobile-Generation Context

Aim

To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and

CDMA, and examine how their assumed user capacity and interference behaviour differ.

Objectives

By the end of the experiment, students should be able to:

1. Model how FDMA, TDMA and CDMA allocate communication resources among

users.

2. Estimate and compare user capacity under stated assumptions.

3. Plot how estimated CDMA SIR changes as the number of simultaneous users increases.

4. Relate the access methods to representative mobile generations and interpret the limits

of the simplified model.

Requirements

• Computer with MATLAB

• No hardware is required

Theoretical Background

FDMA gives users separate frequency channels. TDMA gives users separate time slots. CDMA

allows users to share a frequency band and time by using different spreading codes.

This experiment uses a simplified analytical model to illustrate those differences. It assumes

ideal FDMA and TDMA resource separation, and estimates CDMA SIR using:

$SIR =$

G

$K - 1$

where G is the processing gain and K is the number of simultaneous users. SIR in decibels is:

$SIRdb = 10 \log 10\, SIR$

The model assumes ideal separation for FDMA and TDMA, and equal received power for CDMA users. It does not model fading, power control, adjacent-channel leakage, guard intervals or standard-specific constraints.

Procedure

Parameter	Value used	Meaning
Total bandwidth	1 MHz	Total available frequency bandwidth
FDMA channel width	200 kHz	Bandwidth allocated to each FDMA user
TDMA slots	8	Slots available per frame
CDMA processing gain	64	Assumed spreading/processing gain
CDMA target SIR	7 dB	Minimum assumed SIR for the capacity estimate

Algorithm:

a. Clear the MATLAB workspace, Command Window and open figures.

b. Enter the system parameters.

c. Estimate the FDMA and TDMA user limits.

d. Convert the CDMA target SIR from decibels to a linear value.

e. Estimate the CDMA user limit.

f. Calculate CDMA SIR for a range of simultaneous user counts.

g. Display the capacity table and generate the plots.

h. Compare the results with the assumptions and explain their limitations.

Procedure:

1. Run the MATLAB program below.

2. Record the assumed bandwidth, channel width, number of TDMA slots, CDMA processing gain and target SIR.

3. Inspect the printed capacity table and the capacity and CDMA-SIR plots.

4. Inspect the generation comparison table and the conceptual time-frequency resource

diagrams.

5. Change at least one parameter for each access method and rerun the program.

6. Compare the new and baseline results; explain the impact of the changed assumptions

Program 1 — Capacity estimate for FDMA, TDMA and CDMA

Objective 1: To estimate and compare the illustrative number of users supported by FDMA,

TDMA and CDMA using the specified system parameters.

MATLAB program:

```
>> % Comparison of FDMA, TDMA and CDMA

% Simplified illustrative model; not a full cellular-network simulation.

clear; clc; close all;

% Illustrative system parameters

totalBandwidthHz = 1e6; % Total bandwidth: 1 MHz

fdmaChannelHz = 200e3; % FDMA channel width: 200 kHz

tdmaSlots = 8; % Number of TDMA slots per frame

processingGain = 64; % CDMA processing gain

targetSIRdB = 7; % Minimum acceptable CDMA SIR

% Capacity estimates under the stated assumptions

fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);

tdmaUsers = tdmaSlots;

targetSIR = 10^(targetSIRdB / 10);

cdmaUsers = floor(1 + processingGain / targetSIR);

fprintf('Illustrative maximum supported users\n');

fprintf('FDMA: %d users\n', fdmaUsers);

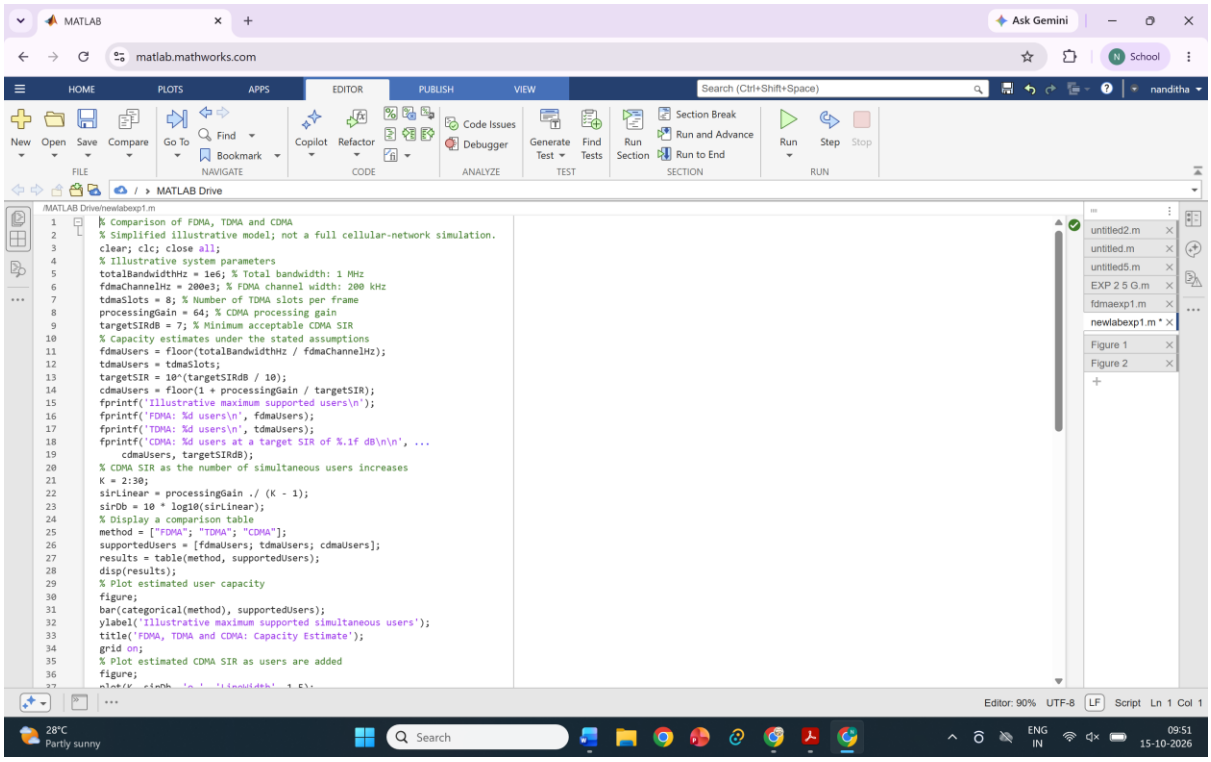
fprintf('TDMA: %d users\n', tdmaUsers);

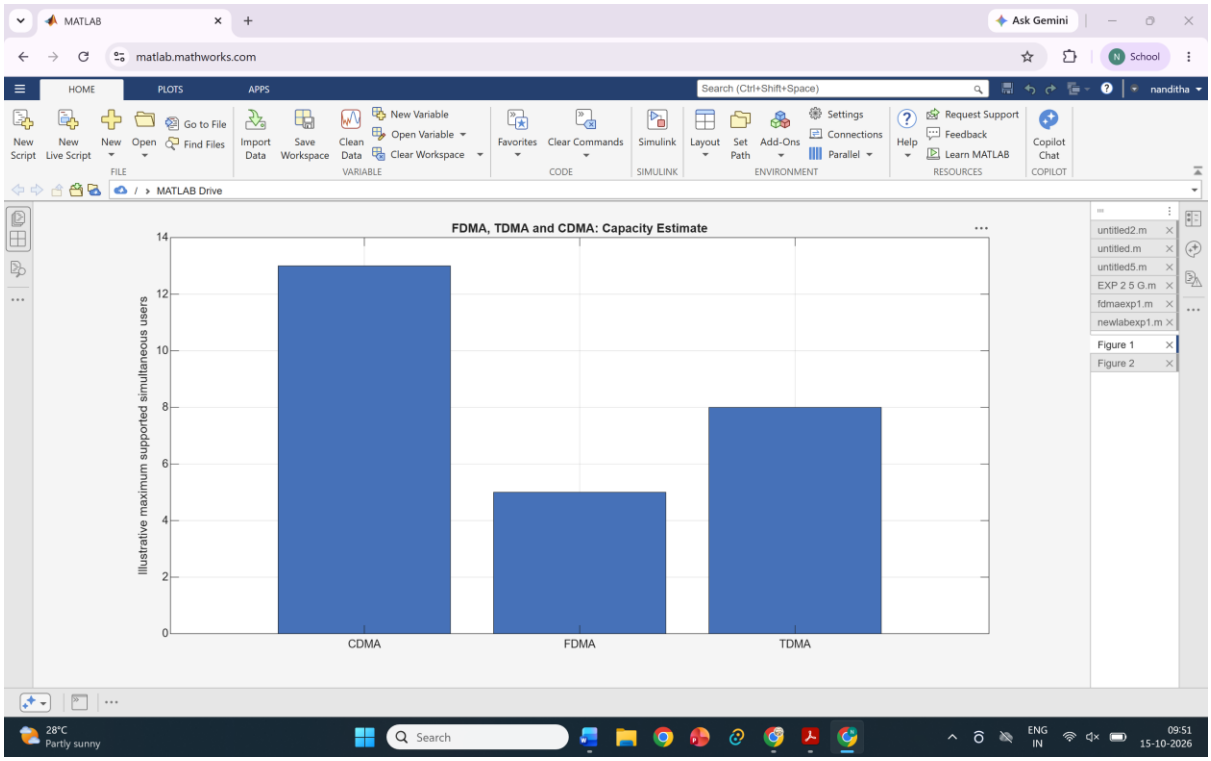
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
```

```
cdmaUsers, targetSIRdB);
% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);
% Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

Result:

Maximum supported simultaneous users

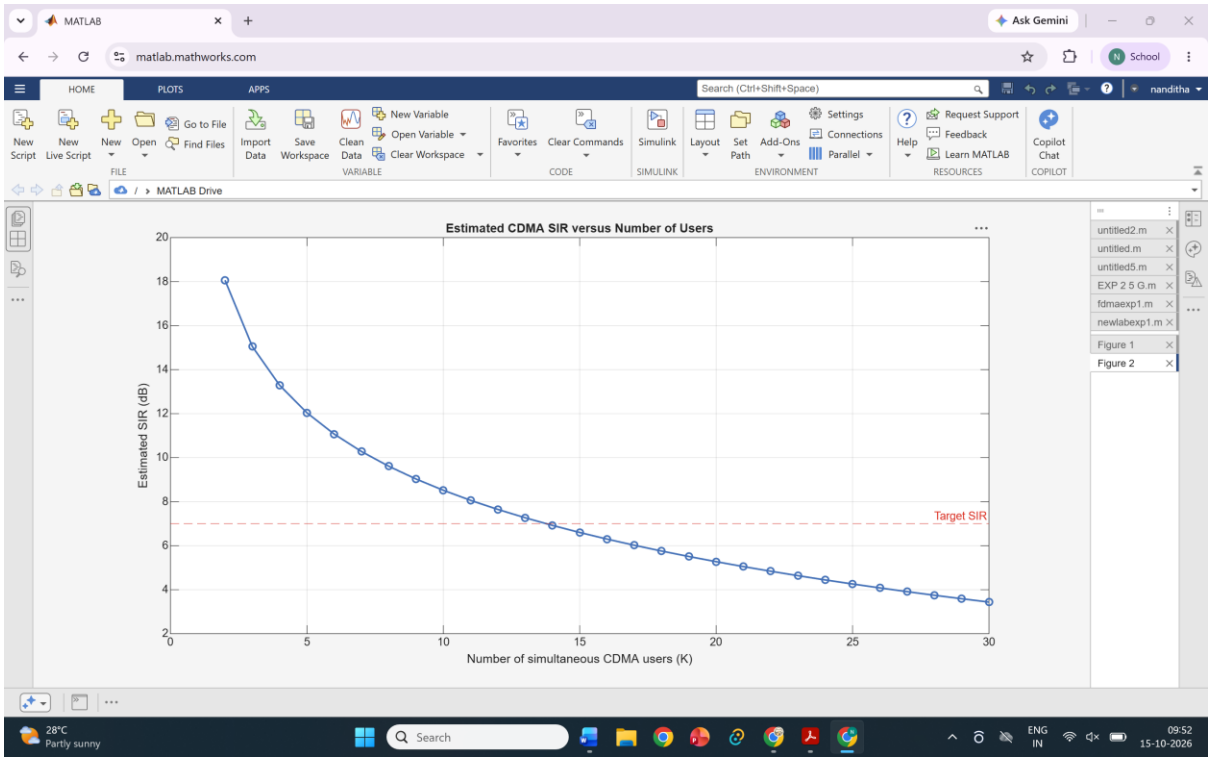




Objective 2: CDMA SIR analysis:

To analyze how the estimated CDMA signal-to-interference ratio changes as the number of simultaneous users increases.

Finding out the SIR for the technique with the highest number of users



Objective 3: Mobile-generation comparison:

To relate representative multiple-access techniques to the evolution of mobile communication

systems from 1G to 5G.

MATLAB Program:

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];

system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];

accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];

keyIdea = ["Separate frequency channels"; ...

"Separate time slots"; ...

"Code-based user separation"; ...

"Wideband code-based access"; ...

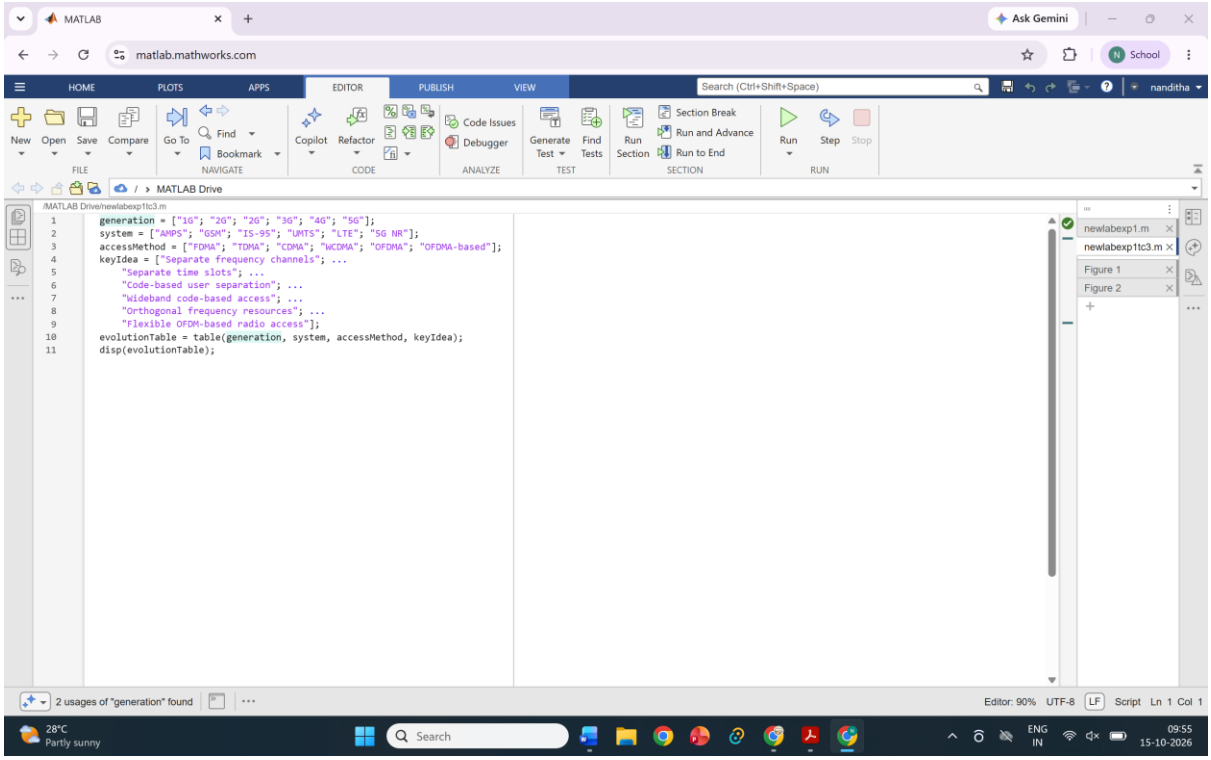
"Orthogonal frequency resources"; ...

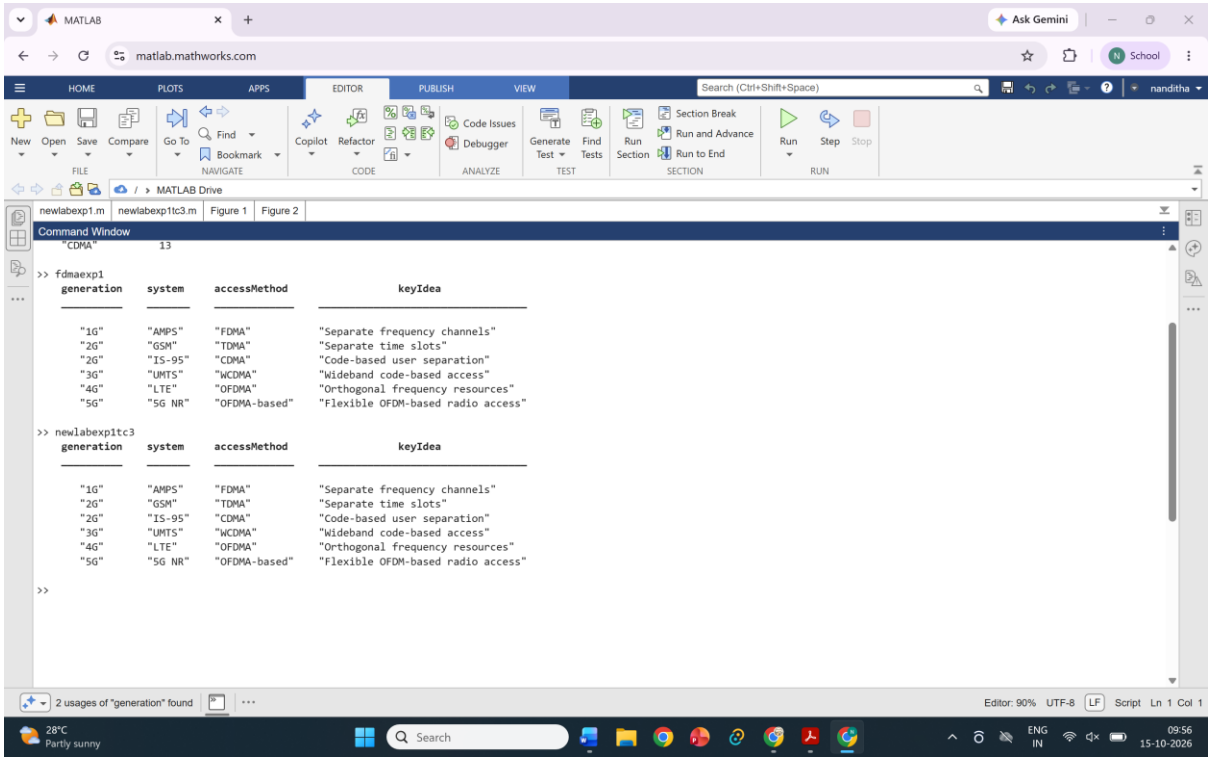
"Flexible OFDM-based radio access"];

evolutionTable = table(generation, system, accessMethod, keyIdea);

disp(evolutionTable);
```

Result:





Objective 4: Resource-allocation diagrams:

To visualize how FDMA, TDMA and CDMA assign frequency, time and code resources to

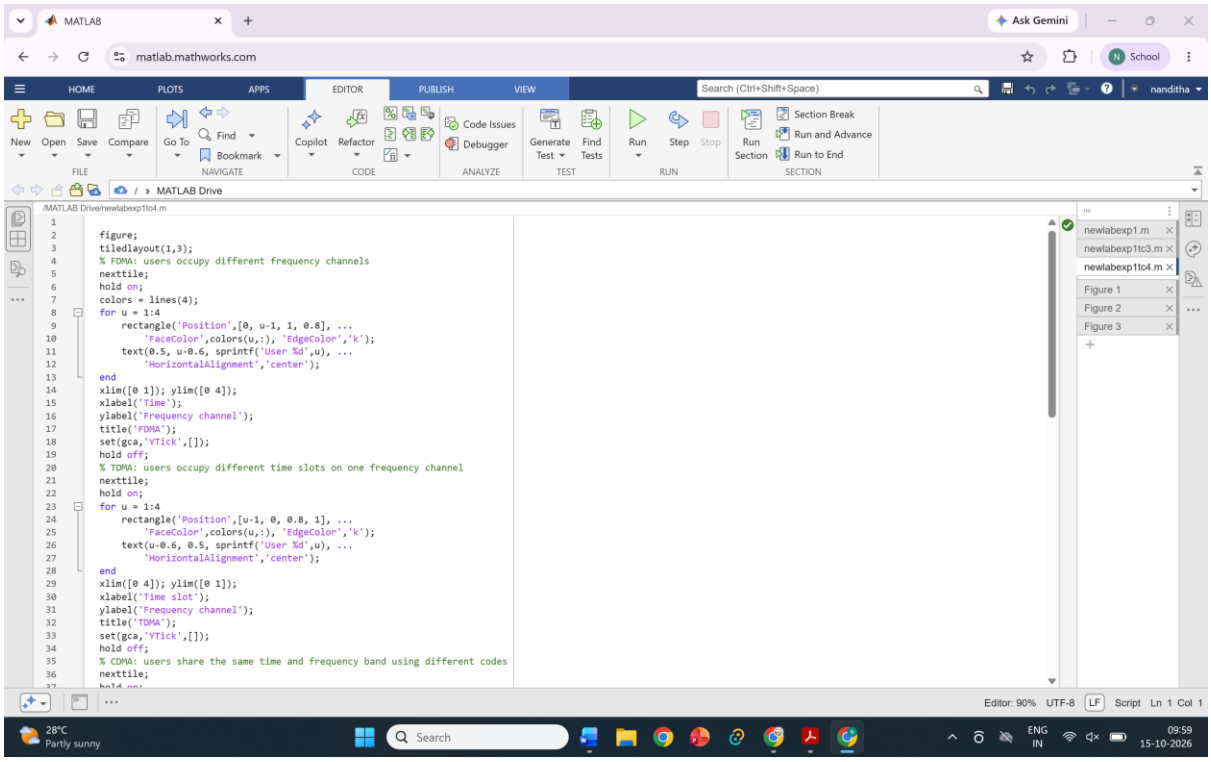
multiple users.

MATLAB Program:

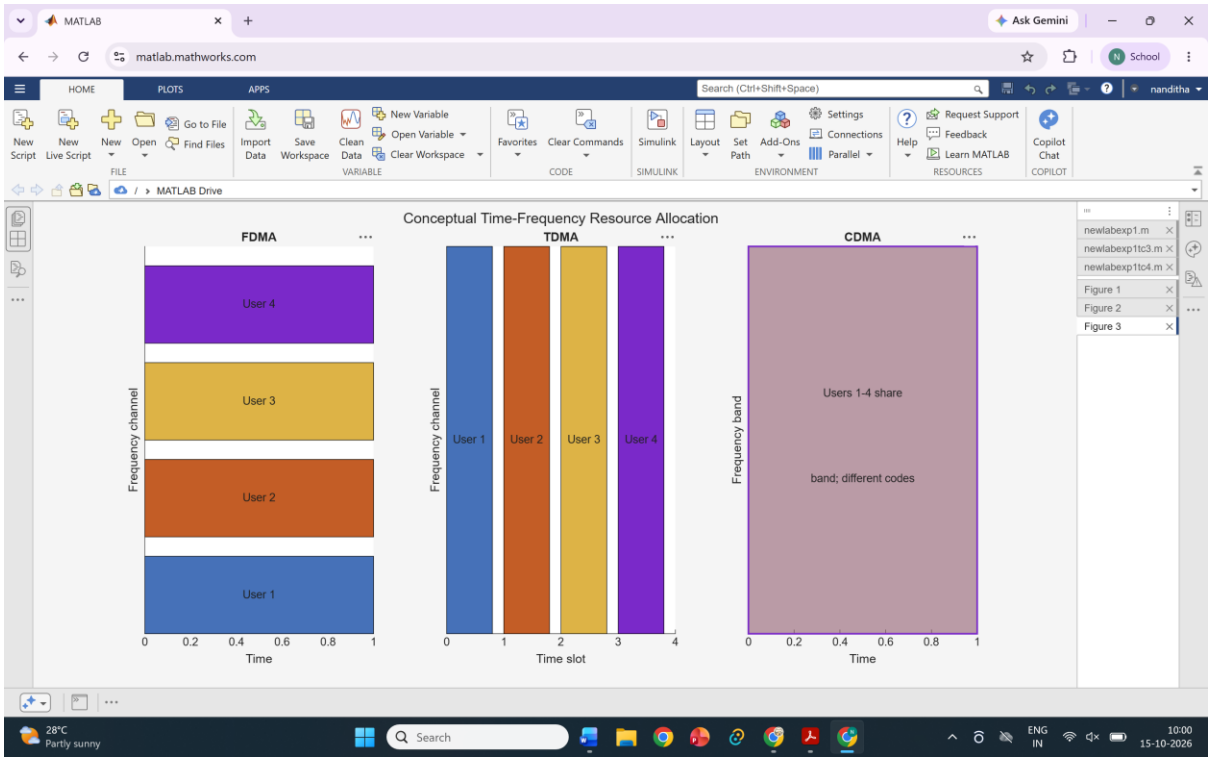
```
Time-frequency resource-allocation diagrams
figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
rectangle('Position',[0, u-1, 1, 0.8], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.5, u-0.6, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
rectangle('Position',[u-1, 0, 0.8, 1], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(u-0.6, 0.5, sprintf('User %d',u), ...
'HorizontalAlignment','center');
```

```
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
hold on;
for u = 1:4
rectangle('Position',[0, 0, 1, 1], ...
'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

Result :



```
figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
    rectangle('Position',[0, u-1, 1, 0.8], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(0.5, u-0.6, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[u-1, 0, 0.8, 1], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(u-0.6, 0.5, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
```



Conclusion

The four MATLAB programs demonstrate how FDMA, TDMA and CDMA allocate communication resources and how the assumed parameters affect their estimated user capacity.

The CDMA analysis shows that estimated SIR decreases as more users share the same

frequency band. The mobile-generation table and resource-allocation diagrams connect these

techniques to the development of wireless communication systems.

These results are based on simplified assumptions. In particular, the model assumes ideal

FDMA and TDMA separation and does not include real-world effects such as fading, power

control, adjacent-channel interference or guard intervals. Therefore, the results illustrate the

operating principles; they should not be treated as a universal performance ranking of the three

methods.

Experiment Number: 2

MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell Network

Aim

To model a simplified hexagonal cellular network and study how cell radius affects coverage,

estimated user capacity and handoff as a mobile crosses a cell boundary.

Objectives

By the end of the experiment, students should be able to:

1. Model adjacent hexagonal cell coverage and identify cell centers and boundaries.

2. Estimate the number of cells required to cover a fixed service area for different cell radii.

3. Compare estimated users served under an explicit per-cell capacity assumption.

4. Simulate a mobile moving across a cell boundary and identify the serving-cell transition.

Requirements

- **Computer with MATLAB R2019b or later.**
- **No RF hardware is required; this is a geometric and analytical simulation.**

Theoretical Background

A regular hexagonal cell is an idealized coverage region. For cell radius R, measured from

the center to a vertex, its area is $A_{cell} = (3\sqrt{3}/2)R^2$. A first-order estimate for a service area

A is $N_{cells} = \text{ceil}(A/A_{cell})$. This area ratio ignores boundary overlap, roads, terrain,

irregular coverage and the fact that real cell layouts extend beyond the service boundary.

For a simple capacity comparison, assume each cell can serve C users at the same time.

Estimated network capacity is $N_{cells} \times C$. This is a planning illustration, not a prediction of

a real network; capacity also depends on spectrum, scheduling, traffic, interference, backhaul

and radio conditions.

For handoff, the serving cell is approximated as the cell center nearest to the mobile. A

handoff is indicated when the nearest-center identity changes. Real systems use measured

radio quality, hysteresis, time-to-trigger, load and network policy, so this geometric rule is

only a teaching model.

Parameters and Assumptions

Parameter	Baseline value	Meaning
Service area	5 km^2	Area to be covered by the idealized layout
Cell radii	1.0 km and 0.5 km	Baseline and split-cell cases
Per-cell simultaneous-user capacity	30 users/cell	Same assumed capacity per cell in each case
Handoff geometry	Two adjacent cells	Mobile follows a straight path across their shared boundary

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Record any values changed. The capacity comparison holds the per-cell capacity constant to

isolate the effect of cell count; it does not imply that smaller real cells always have exactly

this capacity.

Procedure

1. Run the MATLAB script below and record the cell area, estimated cell count and total

illustrative capacity for each radius.

2. Inspect the hexagonal layout and confirm that adjacent cells share edges without gaps in

the idealized tiling.

3. Inspect the mobile-path plot and note the simulated handoff point.

4. Change the service area, radii or users-per-cell assumption one at a time; rerun and

compare results.

5. Explain why the geometric cell count and nearest-center handoff differ from an operational cellular network.

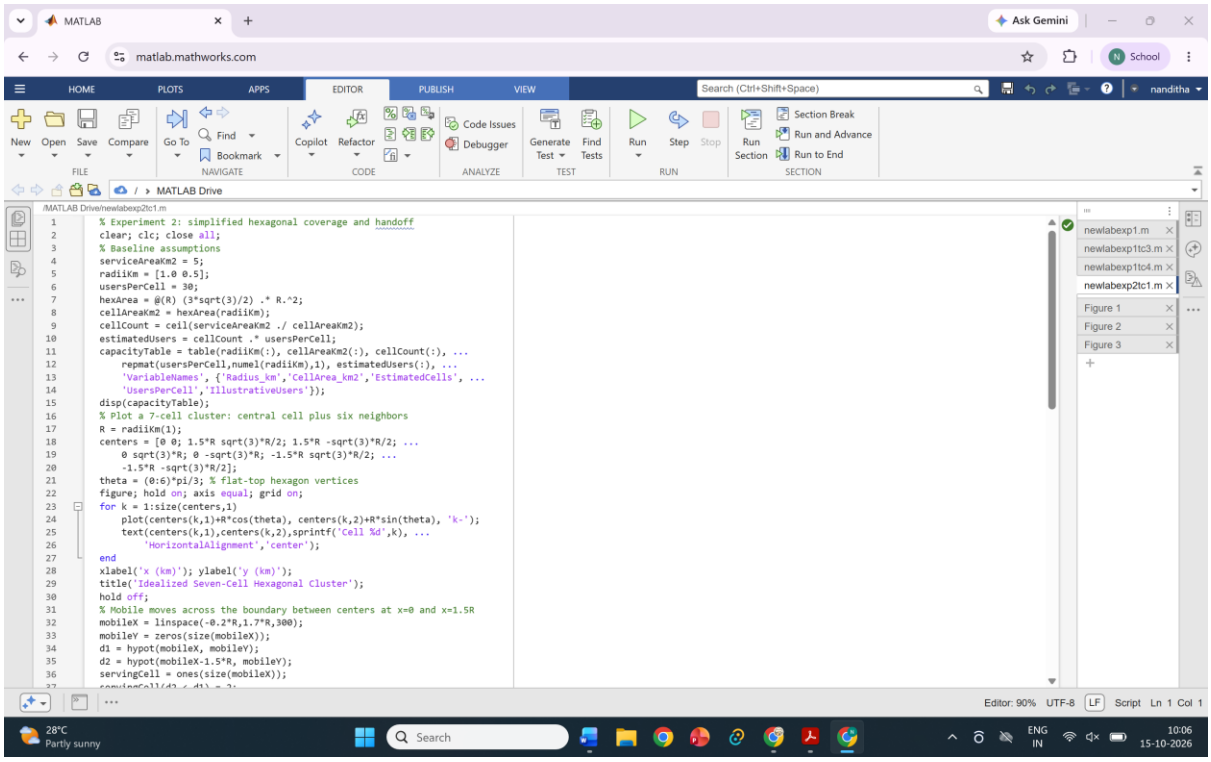
MATLAB Program — Coverage, Cell Splitting and Handoff

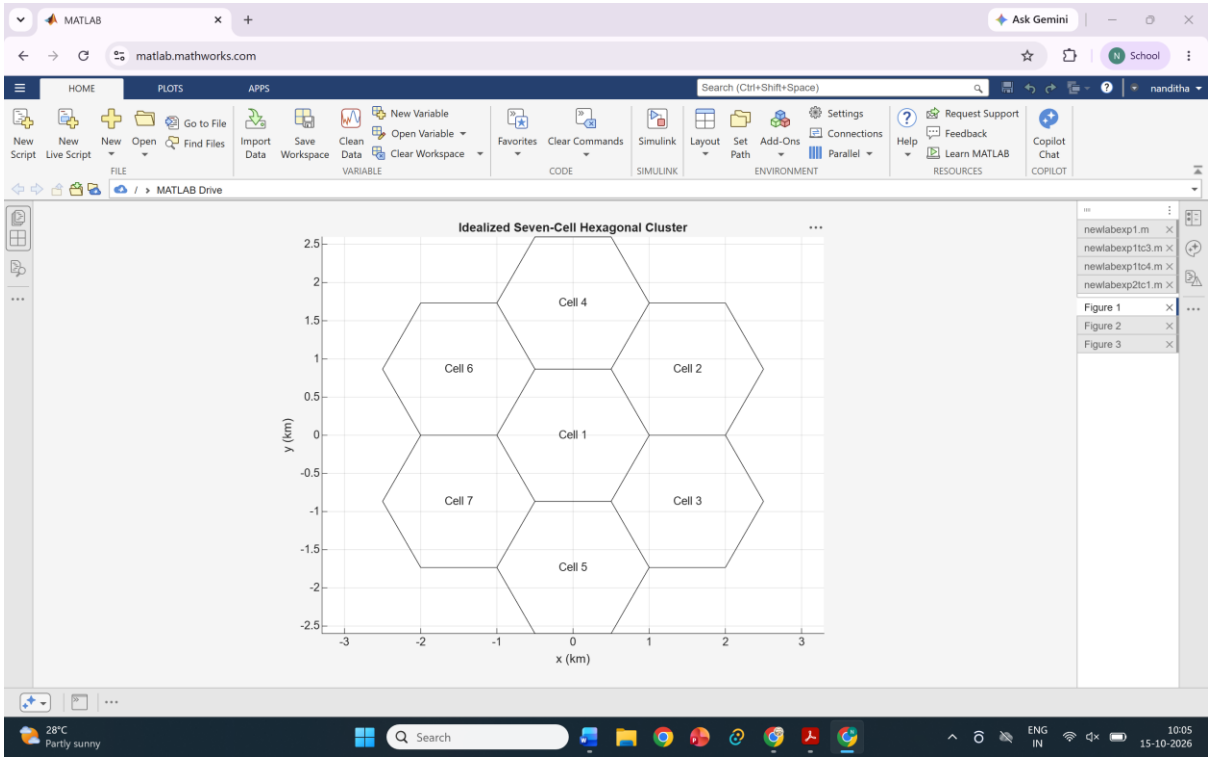
```
% Experiment 2: simplified hexagonal coverage and handoff
clear; clc; close all;
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
-1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3; % flat-top hexagon vertices
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
```

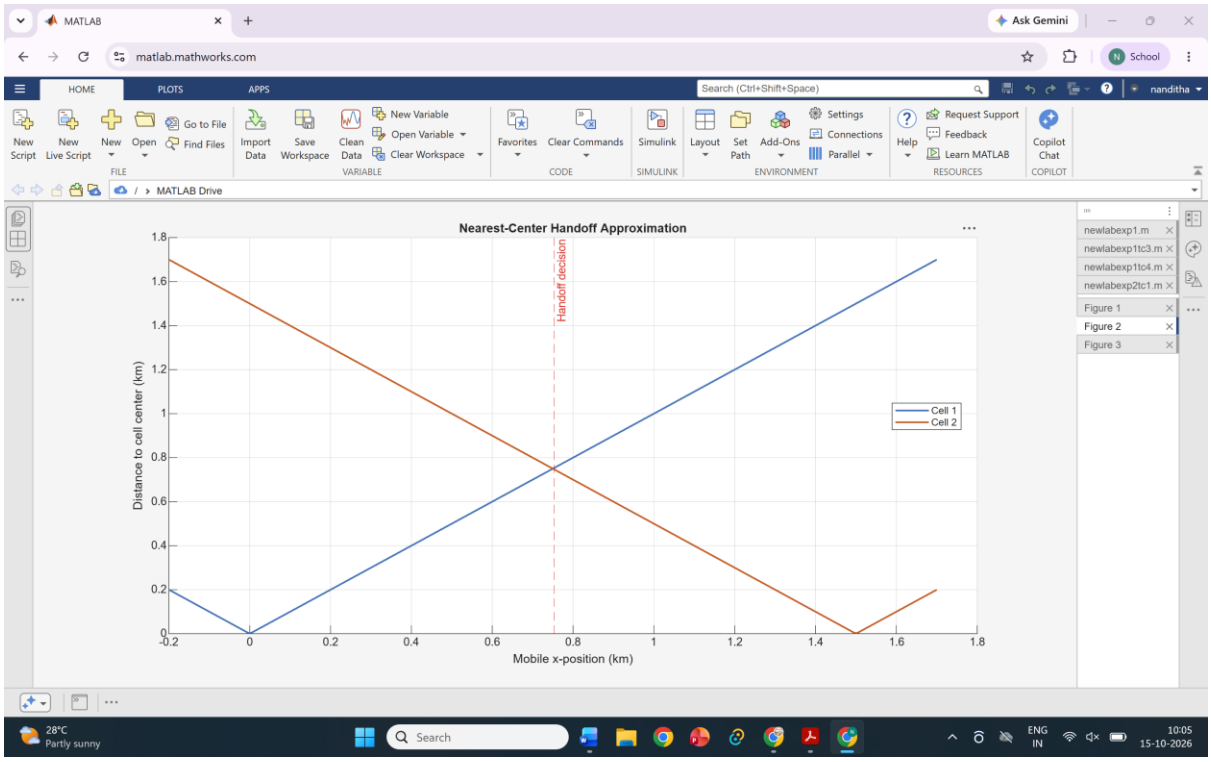
```
'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;
% Mobile moves across the boundary between centers at x=0 and x=1.5R
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell) ~= 0,1,'first') + 1;
figure; hold on; grid on;
plot(mobileX,d1,'LineWidth',1.5);
plot(mobileX,d2,'LineWidth',1.5);
if ~isempty(handoffIndex)
xline(mobileX(handoffIndex),'--r','Handoff decision');
end
xlabel('Mobile x-position (km)'); ylabel('Distance to cell center (km)');
legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');
hold off;
figure; stairs(mobileX,servingCell,'LineWidth',1.5); grid on;
xlabel('Mobile x-position (km)'); ylabel('Serving-cell ID');
title('Serving Cell Along the Mobile Path');
if ~isempty(handoffIndex)
fprintf('Geometric handoff at x = %.3f km\n',mobileX(handoffIndex));
```

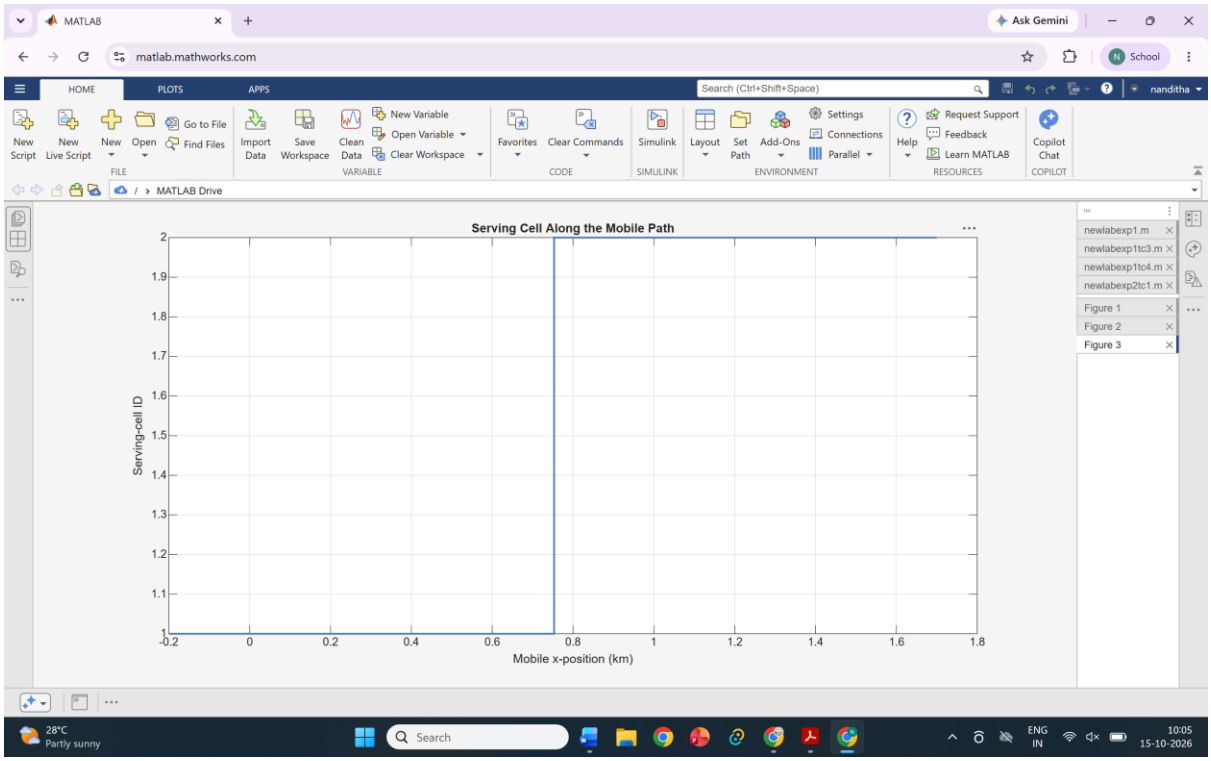
end

Result:


MATLAB
matlab.mathworks.com
Ask Gemini
School
nanditha
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
MATLAB Drive
% Experiment 2: simplified hexagonal coverage and handoff
clear; clc; close all;
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
-1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3; % flat-top hexagon vertices
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;
% Mobile moves across the boundary between centers at x=0 and x=1.5R
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
newlabexp1.m
newlabexp11tc3.m
newlabexp11tc4.m
newlabexp2tc1.m
Figure 1
Figure 2
Figure 3
Editor: 90%
UTF-8
Script
Ln 1 Col 1
28°C
Partly sunny
Search
ENG
IN
10:06
15-10-2026


MATLAB
matlab.mathworks.com
Ask Gemini
School
nanditha
HOME
PLOTS
APPS
MATLAB Drive
Idealized Seven-Cell Hexagonal Cluster
Cell 1
Cell 2
Cell 3
Cell 4
Cell 5
Cell 6
Cell 7
y (km)
x (km)
newlabexp1.m
newlabexp11tc3.m
newlabexp11tc4.m
newlabexp2tc1.m
Figure 1
Figure 2
Figure 3
28°C
Partly sunny
Search
ENG
IN
10:05
15-10-2026





Results to Record

Radius (km)	Cell area (km²)	Estimated cells	Users/cell	Illustrative users
1	2.5981	2	30	60
0.5	0.64952	8	30	240

Observation	Recorded result
Simulated handoff x-position (km)	0.753 km
Effect of halving radius on estimated cells	
Effect of changed parameter	

Analysis Questions

1. Why does halving the radius increase the idealized number of cells by approximately four

for a fixed area?

2. What assumption makes the estimated total user capacity scale directly with the number

of cells?

3. At what position does the nearest-center rule switch the serving cell, and why is that point

the geometric boundary?

4. Name two radio or network factors absent from this model that could shift or delay a real

handoff.

Conclusion

Summarize how cell radius changes the idealized cell count and assumed network capacity,

and report the simulated handoff location. State that the model uses ideal hexagons, an arearatio

count, fixed per-cell capacity and nearest-center handoff; it does not model RF

propagation, overlap, fading, load or handoff timers.

Experiment Number: 3

MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

Aim

To compare a simplified free-space path-loss estimate across representative sub-1 GHz, midband

and mmWave frequencies, and examine how distance and frequency reuse affect

received signal and a simple interference measure.

Objectives

By the end of the experiment, students should be able to:

1. Calculate or simulate path loss at representative sub-1 GHz, mid-band and mmWave

frequencies.

2. Compare estimated received signal strength as frequency and distance change.

3. Examine how cell separation or frequency reuse changes a simplified interference ratio.

4. Relate the results to coverage and spectrum-use considerations across mobile generations,

while stating model limitations.

Requirements

• Computer with MATLAB R2019b or later.

• No RF hardware is required.

Theoretical Background

The free-space path-loss equation in decibels is FSPL = 32.44 + $20\log_{10}$(f_MHz) +

$20\log_{10}$(d_km), where f is carrier frequency and d is separation. With fixed transmit power

and antenna gains, estimated received power is P_r = P_t + G_t + G_r − FSPL. In this

idealized equation, doubling distance adds about 6 dB of loss and doubling frequency adds

about 6 dB of loss at the same distance.

For a simple reuse illustration, let the desired transmitter be at distance R and six equal-power

co-channel interferers be at distance D = √(3N)R, where N is a reuse-cluster size. Under a

power-law path-loss exponent n, approximate SIR = (D/R)^n / 6. This idealized first-tier

model omits shadowing, fading, sector antennas, unequal loads, terrain, adjacent-channel

leakage and real frequency-planning constraints. Because desired and interfering links use the

same frequency, frequency-dependent free-space loss cancels in this simplified SIR ratio.

Parameters and Assumptions

Parameter	Baseline value	Meaning
Carrier bands	0.7, 3.5, 28 GHz	Representative sub-1 GHz, mid-band and mmWave points
Transmit power	30 dBm	Illustrative conducted power; antenna gains set to 0 dBi
Distance range	0.1 to 2 km	Free-space comparison range
Path-loss exponent for reuse model	3.5	Illustrative non-free-space distance exponent
Reuse cluster sizes N	1, 3, 7, 12	Larger N implies greater idealized co-channel separation

These are teaching assumptions, not regulatory limits or recommended deployment settings.

The free-space curve is not an urban, indoor or mmWave channel model.

Procedure

1. Run the script and inspect the path-loss and received-power plots.

2. Record the estimated received power at 0.5 km for each frequency.

3. Inspect the reuse-SIR plot and compare the baseline N values.

4. Change transmit power, distance range, path-loss exponent or reuse sizes; rerun and

record the effect.

5. Explain why frequency reuse changes the simplified co-channel SIR and why the freespace

link budget is not a complete coverage prediction.

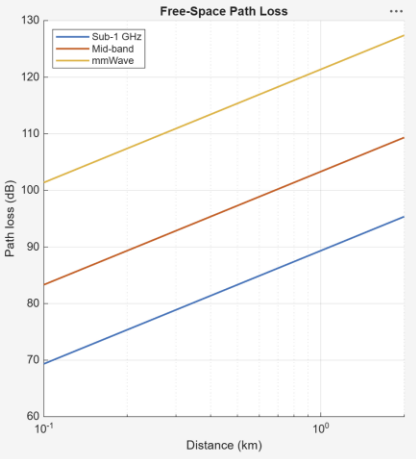
MATLAB Program — Path Loss, Received Power and Reuse Interference

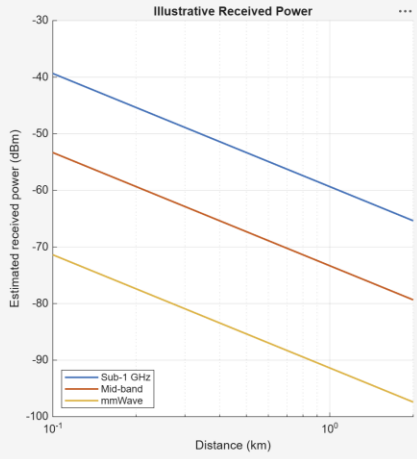
% Experiment 3: illustrative propagation and frequency reuse

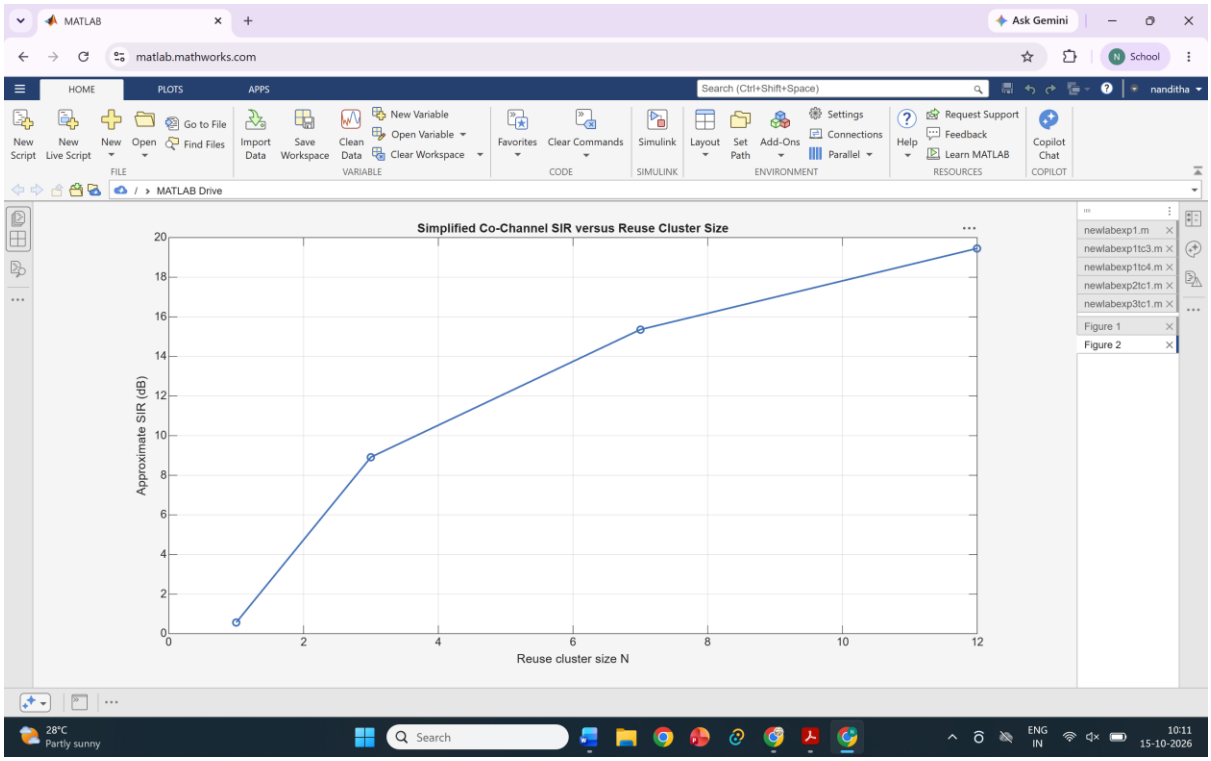
```
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
plot(distanceKm,rx,'LineWidth',1.5);
```

```
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

```
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
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fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
```







Results to Record

Band	Frequency (GHz)	FSPL at 0.5 km (dB)	Estimated Rx (dBm)
Sub-1 GHz	0.7	83.321	-53.321
Mid-band	3.5	97.301	-67.301
mmWave	28	115.36	-85.363

Reuse N	Interferer distance (km)	Approx. SIR (dB)
1	0.86603	0.56811
3	1.5	8.9177
7	2.2913	15.357
12	3	19.454

Analysis Questions

1. At a fixed distance, which representative band has the highest free-space loss? Explain the

frequency term in the equation.

2. How does doubling distance affect FSPL in this model?

3. What trend does the reuse-SIR estimate show as N increases, and which geometric

assumption drives that trend?

4. Why should the calculated received power not be treated as a reliable coverage radius for

an actual cellular system?

5. Relate the bands qualitatively to coverage and capacity trade-offs without treating this

simple model as a comparison of complete mobile generations.

Conclusion

Summarize the frequency and distance trends from the simulation and the effect of reuse

separation on the illustrative SIR. Identify the free-space assumption, fixed transmit power

and idealized six-interferer geometry as limitations. Note that practical coverage depends on

antenna characteristics, environment, bandwidth, receiver sensitivity, interference, regulatory

constraints and the chosen propagation model.